SIlo.c - Code::Blocks 20.03

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<global>

Start here | SIlo.c

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

struct Node* createNode(int data) {
    struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = data;
    newNode->next = NULL;
    return newNode;
}

void insertEnd(struct Node **head, int data) {
    struct Node *newNode = createNode(data);
    if (*head == NULL) {
        *head = newNode;
    } else {
        struct Node *temp = *head;
        while (temp->next != NULL)
            temp = temp->next;
        temp->next = newNode;
    }
}

void display(struct Node *head) {
    if (head == NULL) {
        printf("List is empty\n");
    } else {
        struct Node *temp = head;
        while (temp != NULL) {
            printf("%d ", temp->data);
            temp = temp->next;
        }
        printf("\n");
    }
}
```

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```
40
41  void sortList(struct Node *head) {
42      if (head == NULL) return;
43      struct Node *current = head, *index = NULL;
44      int temp;
45
46      while (current != NULL) {
47          index = current->next;
48          while (index != NULL) {
49              if (current->data > index->data) {
50                  temp = current->data;
51                  current->data = index->data;
52                  index->data = temp;
53              }
54              index = index->next;
55          }
56          current = current->next;
57      }
58  }
59
60  struct Node* reverseList(struct Node *head) {
61      struct Node *prev = NULL, *current = head, *next = NULL;
62
63      while (current != NULL) {
64          next = current->next;
65          current->next = prev;
66          prev = current;
67          current = next;
68      }
69      return prev;
70  }
71
72  struct Node* concatenate(struct Node *head1, struct Node *head2) {
73      if (head1 == NULL) return head2;
74      struct Node *temp = head1;
75      while (temp->next != NULL)
76          temp = temp->next;
77      temp->next = head2;
78      return head1;
```

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```
79  }
80
81  int main() {
82      struct Node *list1 = NULL, *list2 = NULL;
83      int choice, value, list_choice;
84
85      while (1) {
86          printf("\nMenu:\n");
87          printf("1. Insert into List 1\n");
88          printf("2. Insert into List 2\n");
89          printf("3. Display Lists\n");
90          printf("4. Sort List\n");
91          printf("5. Reverse List\n");
92          printf("6. Concatenate Lists\n");
93          printf("7. Exit\n");
94          printf("Enter your choice: ");
95          scanf("%d", &choice);
96
97          switch (choice) {
98              case 1:
99                  printf("Enter value to insert in List 1: ");
100                 scanf("%d", &value);
101                 insertEnd(&list1, value);
102                 break;
103             case 2:
104                 printf("Enter value to insert in List 2: ");
105                 scanf("%d", &value);
106                 insertEnd(&list2, value);
107                 break;
108             case 3:
109                 printf("List 1: ");
110                 display(list1);
111                 printf("List 2: ");
112                 display(list2);
113                 break;
114             case 4:
115                 printf("Sort which list (1 or 2)? ");
116                 scanf("%d", &list_choice);
117                 if (list_choice == 1)
```

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```
105                 scanf("%d", &value);
106                 insertEnd(&list2, value);
107                 break;
108             case 3:
109                 printf("List 1: ");
110                 display(list1);
111                 printf("List 2: ");
112                 display(list2);
113                 break;
114             case 4:
115                 printf("Sort which list (1 or 2)? ");
116                 scanf("%d", &list_choice);
117                 if (list_choice == 1)
118                     sortList(list1);
119                 else
120                     sortList(list2);
121                 break;
122             case 5:
123                 printf("Reverse which list (1 or 2)? ");
124                 scanf("%d", &list_choice);
125                 if (list_choice == 1)
126                     list1 = reverseList(list1);
127                 else
128                     list2 = reverseList(list2);
129                 break;
130             case 6:
131                 list1 = concatenate(list1, list2);
132                 list2 = NULL;
133                 printf("Lists concatenated. List 1 now contains both lists.\n");
134                 break;
135             case 7:
136                 exit(0);
137             default:
138                 printf("Invalid choice. Try again.\n");
139         }
140     }
141     return 0;
142 }
143
```

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```
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Menu:
1. Insert into List 1
2. Insert into List 2
3. Display Lists
4. Sort List
5. Reverse List
6. Concatenate Lists
7. Exit
Enter your choice: 1
Enter value to insert in List 1: 45

Menu:
1. Insert into List 1
2. Insert into List 2
3. Display Lists
4. Sort List
5. Reverse List
6. Concatenate Lists
7. Exit
Enter your choice: 1
Enter value to insert in List 1: 34

Menu:
1. Insert into List 1
2. Insert into List 2
3. Display Lists
4. Sort List
5. Reverse List
6. Concatenate Lists
7. Exit
Enter your choice: 2
Enter value to insert in List 2: 345

Menu:
1. Insert into List 1
2. Insert into List 2
3. Display Lists
4. Sort List
5. Reverse List
6. Concatenate Lists
```

```
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7. Exit
Enter your choice: 2
Enter value to insert in List 2: 36

Menu:
1. Insert into List 1
2. Insert into List 2
3. Display Lists
4. Sort List
5. Reverse List
6. Concatenate Lists
7. Exit
Enter your choice: 2
Enter value to insert in List 2: 156

Menu:
1. Insert into List 1
2. Insert into List 2
3. Display Lists
4. Sort List
5. Reverse List
6. Concatenate Lists
7. Exit
Enter your choice: 3
List 1: 45 34
List 2: 345 36 156

Menu:
1. Insert into List 1
2. Insert into List 2
3. Display Lists
4. Sort List
5. Reverse List
6. Concatenate Lists
7. Exit
Enter your choice: 5
Reverse which list (1 or 2)? 2

Menu:
1. Insert into List 1
2. Insert into List 2
```

```
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5. Reverse List
6. Concatenate Lists
7. Exit
Enter your choice: 6
Lists concatenated. List 1 now contains both lists.

Menu:
1. Insert into List 1
2. Insert into List 2
3. Display Lists
4. Sort List
5. Reverse List
6. Concatenate Lists
7. Exit
Enter your choice: 4
Sort which list (1 or 2)? 1

Menu:
1. Insert into List 1
2. Insert into List 2
3. Display Lists
4. Sort List
5. Reverse List
6. Concatenate Lists
7. Exit
Enter your choice: 3
List 1: 34 36 45 156 345
List 2: List is empty

Menu:
1. Insert into List 1
2. Insert into List 2
3. Display Lists
4. Sort List
5. Reverse List
6. Concatenate Lists
7. Exit
Enter your choice: 7

Process returned 0 (0x0)   execution time : 94.358 s
Press any key to continue.
```